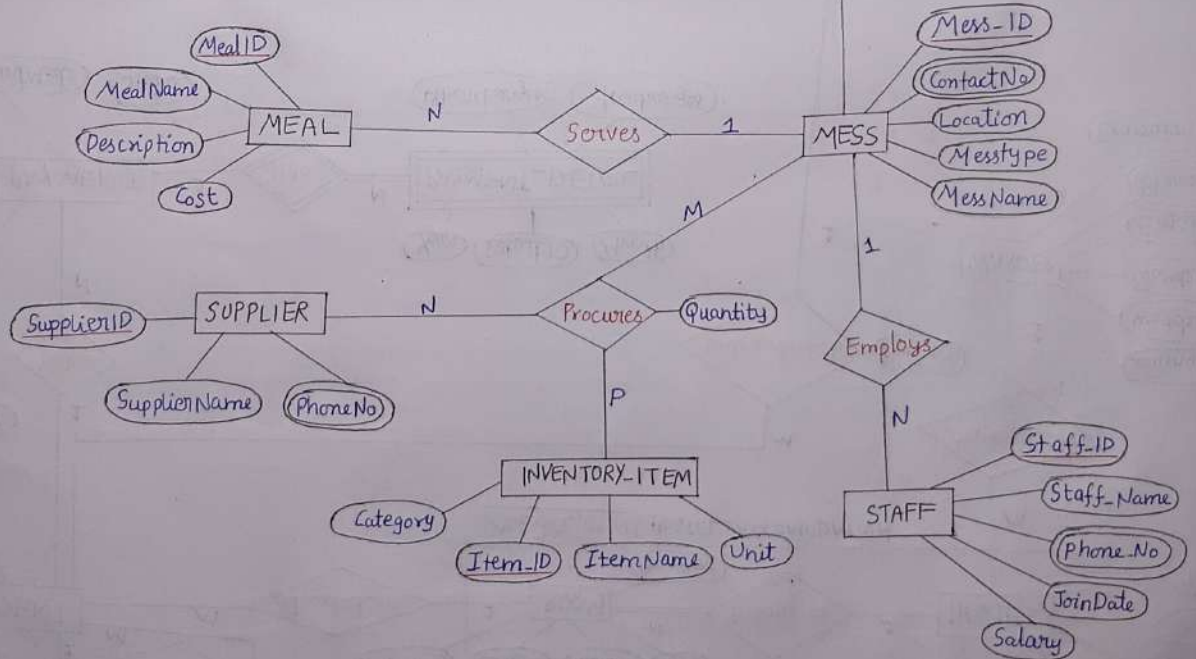
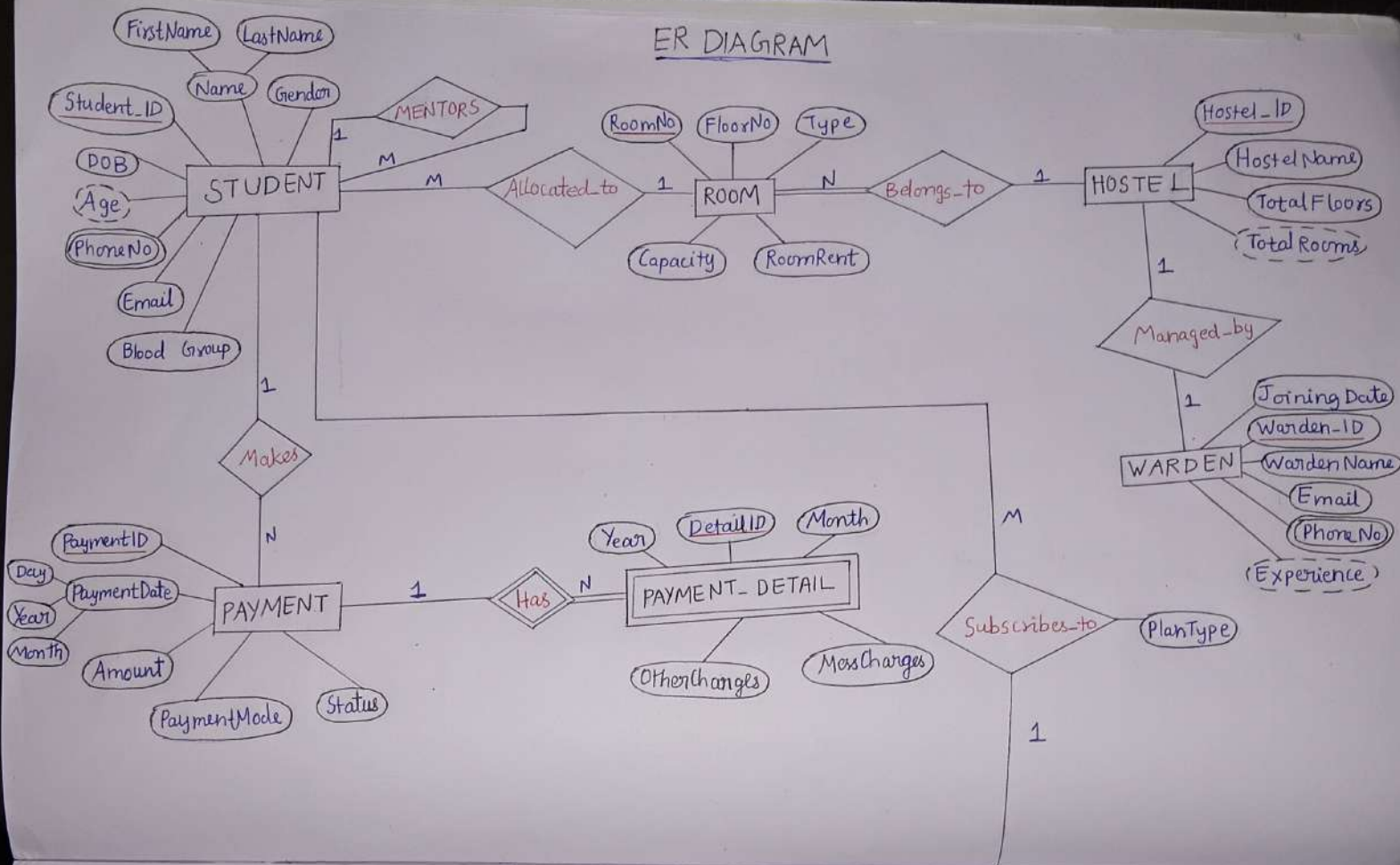


ER DIAGRAM



HOSTEL AND MESS MANAGEMENT

ER-DIGRAM:

✓ Relationships:

1. Student mentors Student (1:N)
2. Student is allocated to Room (M:1)
3. Room belongs to Hostel (N:1)
4. Hostel is managed by Warden (1:1)
5. Student makes Payment (1:N)
6. Payment has Payment Detail (1:N)
7. Student subscribes to Mess (1:N)
8. Mess serves Meal (1:N)
9. Supplier produces Inventory Item (N:M:P)
10. Mess employs Staff (1:N)

Unary Relation: Student Mentors Student (1:N)

Ternary Relation: Mess procures Inventory-Item from Supplier (N:M:P)

✓ Entities:

Strong Entity: Student, Hostel, Warden, Room, Mess, Staff, Supplier, Inventory Item, Meal, Payment

Weak Entity: Payment-Detail

✓ Attributes:

Key attribute: StudentID, HostelID, WardenID, RoomNo, MessID, StaffID, SupplierID, ItemID, MealID, PaymentID.

Partial Attribute: DetailID

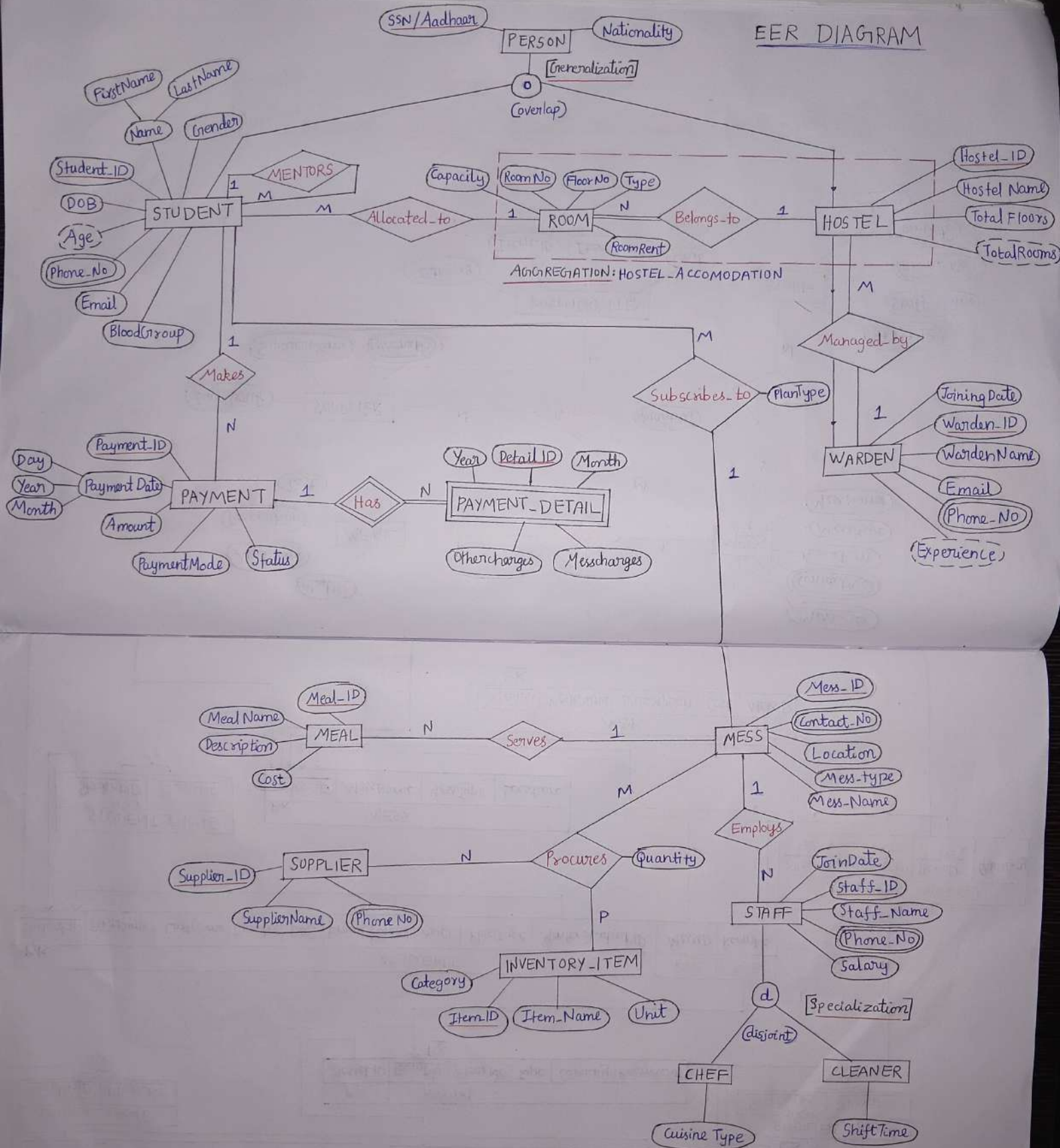
Composite Attribute: Name, Payment Date

Multivalued Attribute: PhoneNo

Derived Attribute: Age

Simple Attribute: Rest all attribute.

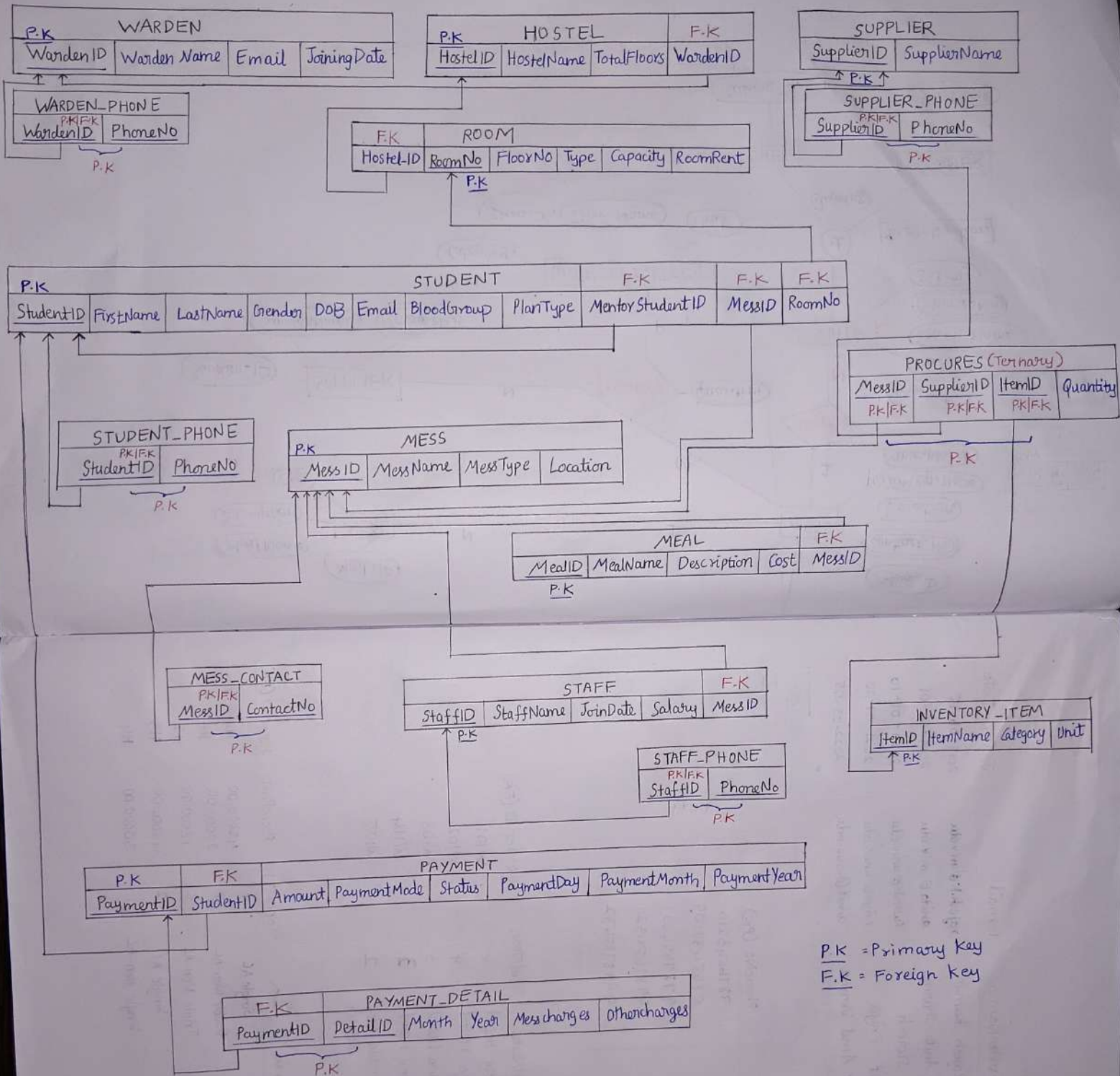
EER DIAGRAM



EER DIAGRAM

- Generalization: Person is generalized into student and Warden.
- Specialization: Staff is specialized into chef and cleaner based on specific roles.
- Aggregation: The Hostel-Room Accomodation relationship is treated as a higher level aggregated entity allocated to student.
- Disjoint specialization (d): A staff member can belong to either chef or cleaner, not both.
- Overlapping (o): A person can belong to more than one subclass simultaneously.

RELATIONAL SCHEMA



P.K = Primary key
F.K = Foreign key

TABLES

1. WARDEN

WardenID(PK)	WardenName	Email	Joining Date
W101	Dr. Rajesh Kumar	rajesh.k@univ.edu	2018-06-15
W102	Prof. Anita Sharma	anita@univ.edu	2019-08-01
W103	Dr. Suresh	suresh@univ.edu	2020-01-10
W104	Prof. Priya	priya@univ.edu	2021-07-20
W105	Dr. Amit Verma	amit@univ.edu	2022-03-05

2. WARDEN_PHONE

WardenID (PK, FK)	PhoneNo (PK)
W101	9876543210
W101	8765432109
W102	9787654321
W103	9987654321
W104	9998765432

3. HOSTEL

HostelID (PK)	HostelName	Total Floors	WardenID (FK)
H1	Cauvery Hostel	4	W101
H2	Ganga Hostel	4	W102
H3	Yamuna Hostel	3	W103
H4	Krishna Hostel	3	W104
H5	Narmada PG Hostel	2	W105

4. ROOM

RoomNo (PK)	Floor No	Type	Capacity	RoomRent	HostelID (FK)
R101	1	Single AC	1	45000.00	H1
R102	1	Double Non-AC	2	25000.00	H1
R201	2	Triple Non-AC	3	18000.00	H2
R202	2	Single AC	2	45000.00	H3
R301	3	Single Non-AC	1	30000.00	H4

5. STUDENT

StudentID (PK)	FirstName	Last Name	Gender	DOB	BloodGroup
S1001	Rahul	Sharma	Male	2003-4-12	B+
S1002	Sneha	Patel	Female	2004-8-25	O+
S1003	Rohan	Verma	Male	2005-1-15	A+
S1004	Ananya	Iyer	Female	2005-9-30	AB+
S1005	Vikram	Singh	Male	2004-11-5	B-

Email	PlanType	MentorStudentID(FK)	RoomNo(FK)	MessID(FK)
rahul.s@univ.edu	Premium Veg	NULL	R101	M1
sneha.p@univ.edu	Standard Non-Veg	NULL	R201	M2
rohan.v@univ.edu	Standard Veg	S1001	R102	M1
ananya.i@univ.edu	Standard Veg	S1002	R201	M2
vikram.s@univ.edu	Premium Non-Veg	S1001	R202	M3

6. STUDENT-PHONE

StudentID (PK, FK)	Phone No (PK)
S1001	9876500000
S1001	9876540000
S1002	9876543000
S1003	9876543200
S1004	9876543210

7. MESS

MessID (PK)	MessName	Mess Type	Location
M1	North Dining Hall	Pure Veg	North Campus Block A
M2	South Dining Hall	Non-Veg Special	South Campus Block B
M3	Central Mess	Mixed	Central Plaza Block C
M4	Executive Dining	Continental	Guest House Block D
M5	Fast Food & Snacks	Quick Bites	Student Activity Center

8. MESS_CONTACT

MessID (PK, FK)	Contact No (PK)
M1	044-220011
M1	044-220012
M2	044-220021
M3	044-220031
M4	044-220041

9. MEAL

MealID(PK)	MealName	Cost	Description
ML1	North Indian Thali	120.00	Roti, Dhal Makhani, Paneer, Rice
ML2	South Indian Meals	90.00	Sambar, Rasam, Curd, Rice, Poriyal
ML3	Chicken Biryani	180.00	Chicke Dum Biryani, Raita
ML4	Continental Breakfast	150.00	Toast, Omlette, Juice, Loffee
ML5	Mini Executive Thali	110.00	Chapati, Mixed Veg Curry, Pulao

10. STAFF

StaffID (PK)	StaffName	JoinDate	Salary
ST1	Ramesh	2019-02-01	35000.00
ST2	Murugan	2020-05-15	28000.00
ST3	Joseph	2018-11-10	38000.00
ST4	Sunita	2021-08-01	18000.00
ST5	Govind	2022-01-20	22000.00

11. STAFF_PHONE

StaffID (PK, FK)	Phone No (PK)
ST1	9444100001
ST2	9444100002
ST3	9444100003
ST3	9444100004
ST4	9444100005

12. SUPPLIER

SupplierID (PK)	SupplierName
SUP1	Fresh Farm Provisions
SUP2	Metro Dairy Products
SUP3	Coastal Poultry & Meats
SUP4	National Grain Traders
SUP5	Green Valley Vegetables

13. SUPPLIER - PHONE

SupplierID (PK,FK)	PhoneNo (PK)
SUP1	9884011111
SUP2	9884022222
SUP3	9884033333
SUP3	9884044444
SUP4	9884055555

14. INVENTORY - ITEM

ItemID (PK)	ItemName	Category	Unit
IT101	Basmati Rice	Grains	Kg
IT102	Toned Milk	Dairy	Litres
IT103	Refined Sunflower Oil	Cooking Oil	Litre
IT104	Fresh Broiler Chicken	Meat	Kg
IT105	Potato	Vegetables	Kg

15. PROCURES

MessID (PK,FK)	SupplierID (PK,FK)	ItemID (PK,FK)	Quantity
M1	SUP4	IT101	500
M1	SUP2	IT102	200
M2	SUP3	IT104	150
M2	SUP1	IT105	300
M3	SUP4	IT103	100

16. PAYMENT

PaymentID(PK)	StudentID(FK)	Amount	Payment Mode	Status
PAY1001	S1001	12500.00	Net Banking	Successful
PAY1002	S1002	14000.00	UPI	Successful
PAY1003	S1003	11000.00	Credit Card	Successful
PAY1004	S1004	11000.00	UPI	Pending
PAY1005	S1005	15000.00	Debit Card	Successful

PaymentDay	Payment Month	PaymentYear
5	8	2026
6	8	2026
7	8	2026
10	8	2026
12	8	2026

17. PAYMENT_DETAIL

PaymentID(PK,FK)	DetailID(PK)	Month	Year	Mess Charges	Other Charges
PAY1001	D1	August	2026	11500.00	1000.00
PAY1002	D1	August	2026	13000.00	1000.00
PAY1003	D1	August	2026	10000.00	1000.00
PAY1004	D2	July	2026	0.00	1000.00
PAY1005	D1	August	2026	13500.00	1500.00

NORMALIZATION:

- First Normal Form (1NF): Every attribute must hold strictly atomic (indivisible) values. Multi-valued attributes and repeating groups are forbidden. A unique primary key must exist for every record.
- Second Normal Form (2NF): The relation must be in 1NF and contain no partial dependencies. Every non-prime attribute must be fully dependent on the entire primary key.

- Third Normal Form (3NF): The relation must be in 2NF and contain no transitive dependencies. Non-prime attributes must not depend on other non-prime attributes. For every functional dependency $X \rightarrow A$, either X is a Super key or A is a Prime Attribute.
- Boyce - Codd Normal Form (BCNF): The relation must be in 3NF. For every non-trivial functional dependency $X \rightarrow A$, X must be a super key.

Table 1: WARDEN

- Schema: WARDEN (WardenID, WardenName, Email, Joining Date)
- Candidate Keys: WardenID, Email
- Functional dependencies: $\text{WardenID} \rightarrow \{\text{WardenName}, \text{Email}, \text{Joining Date}\}$
 $\text{Email} \rightarrow \text{WardenID}$
- Normalization Check:

INF
 2NF
 3NF
 BCNF

} Satisfied, No violation.

Table 2: WARDEN - PHONE

- Schema: WARDEN - PHONE (WardenID, PhoneNo)
- Candidate Keys: (WardenID, PhoneNo)
- Functional Dependencies: $(\text{WardenID}, \text{PhoneNo}) \rightarrow \phi$
- Normalization check:

INF ✓
 2NF ✓
 3NF ✓
 BCNF ✓
 Satisfied, No violation.

Table 3: HOSTEL

* Schema: HOSTEL (HostelID, HostelName, Total Floors, WardenID)

* Candidate Keys: HostelID

* Functional Dependencies: $\text{HostelID} \rightarrow \{ \text{HostelName}, \text{Total Floors}, \text{WardenID} \}$

* Normalization Check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation.

Table 4: ROOM

* Schema: ROOM (RoomNo, FloorNo, Type, Capacity, RoomRent, HostelID)

* Candidate Key: RoomNo.

* Functional Dependencies:

$\text{RoomNo} \rightarrow \{ \text{FloorNo}, \text{Type}, \text{HostelID} \}$

$\text{Type} \rightarrow \{ \text{Capacity}, \text{RoomRent} \}$

* Normalization Check:

1NF ✓

2NF ✓

3NF x } Violation

BCNF x }

$\text{RoomNo} \rightarrow \text{Type}$ and $\text{Type} \rightarrow (\text{Capacity}, \text{RoomRent})$.

Capacity and RoomRent transitively depend on non-key attribute Type. Type is not a Super key, violating 3NF and BCNF.

* Decomposition:

1. ROOM_TYPE (Type, Capacity, RoomRent)

2. ROOM (RoomNo, FloorNo, Type, HostelID)

3NF ✓

BCNF ✓

Table 5: STUDENT

- Schema: STUDENT(StudentID, FirstName, LastName, Gender, DOB, Email, BloodGroup, PlanType, MentorStudentID, RoomNo, MessID)
- Candidate keys: StudentID, Email
- Functional Dependencies:

StudentID $\rightarrow$ { First Name, Last Name, Gender, DOB, Email, BloodGroup, PlanType, MentorStudentID, RoomNo, MessID }

Email $\rightarrow$ StudentID

- Normalization Check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation

Table 6: STUDENT_PHONE

- Schema: STUDENT_PHONE(StudentID, PhoneNo)
- Candidate Keys: (StudentID, PhoneNo)
- Functional Dependencies: (StudentID, PhoneNo) $\rightarrow \phi$
- Normalization check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation.

Table 7: MESS

- Schema: MESS(MessID, MessName, MessType, Location)
- Candidate Key: MessID
- Function Dependencies: MessID $\rightarrow$ { MessName, MessType, Location }
- Normalization Check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation

Table 8: MESS_CONTACT

- Schema: $MESS_CONTACT(\underline{MessID}, \underline{ContactNo})$
- Candidate keys: $(MessID, ContactNo)$
- Functional Dependencies: $(MessID, ContactNo) \rightarrow \phi$
- Normalization check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation

Table 9: MEAL

- Schema: $MEAL(\underline{MealID}, \underline{MealName}, \underline{Description}, \underline{Cost}, \underline{MessID})$
- Candidate key: $MealID$
- Functional Dependency:
 $MealID \rightarrow \{MealName, Description, Cost, MessID\}$
- Normalization check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation.

Table 10: STAFF

- Schema: $STAFF(\underline{StaffID}, \underline{StaffName}, \underline{JoinDate}, \underline{Salary}, \underline{Role}, \underline{MessID})$
- Candidate key: $StaffID$
- Functional Dependency:
 $StaffID \rightarrow \{StaffName, JoinDate, Salary, Role, MessID\}$
- Normalization check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation.

Table 11: STAFF_PHONE

- Schema: STAFF_PHONE (StaffID, PhoneNo)
- Candidate Key: (StaffID, PhoneNo)
- Functional Dependency: (StaffID, PhoneNo) $\rightarrow \phi$
- Normalization check:

1NF ✓
2NF ✓
3NF ✓
BCNF ✓

Satisfied, No violation.

Table 12: SUPPLIER

- Schema: SUPPLIER (SupplierID, SupplierName)
- Candidate Keys: SupplierID
- Functional Dependency: SupplierID $\rightarrow$ SupplierName
- Normalization Check:

1NF ✓
2NF ✓
3NF ✓
BCNF ✓

Satisfied, No violation.

Table 13: SUPPLIER_PHONE

- Schema: SUPPLIER_PHONE (SupplierID, PhoneNo)
- Candidate Keys: (SupplierID, PhoneNo)
- Functional Dependency: (SupplierID, PhoneNo) $\rightarrow \phi$
- Normalization check:

1NF ✓
2NF ✓
3NF ✓
BCNF ✓

Satisfied, No violation

Table 14: INVENTORY - ITEM

- Schema: INVENTORY_ITEM (ItemID, ItemName, Category, Unit)
- Candidate key: ItemID
- Functional Dependency: $\text{ItemID} \rightarrow \{\text{ItemName}, \text{Category}, \text{Unit}\}$
- Normalization check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No violation.

Table 15: PROCURES

- Schema: PROCURES (MessID, SupplierID, ItemID, Quantity)
- Candidate key: (MessID, SupplierID, ItemID)
- Functional Dependency: $(\text{MessID}, \text{SupplierID}, \text{ItemID}) \rightarrow \text{Quantity}$
- Normalization check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No Violation

Table 16: PAYMENT

- Schema: PAYMENT (PaymentID, StudentID, Amount, PaymentMode, Status, PaymentDay, PaymentMonth, PaymentYear)
- Candidate key: PaymentID
- Functional Dependency:
 $\text{PaymentID} \rightarrow \{\text{StudentID}, \text{Amount}, \text{PaymentMode}, \text{Status}, \text{PaymentDay}, \text{PaymentMonth}, \text{PaymentYear}\}$
- Normalization check:

1NF ✓

2NF ✓

3NF ✓

BCNF ✓

Satisfied, No Violation

Table 17: PAYMENT - DETAIL

- Schema: PAYMENT-DETAIL (PaymentID, DetailID, Month, Year, MessCharges, OtherCharges)
 - Candidate Key: (PaymentID, DetailID)
 - Functional Dependency:
(PaymentID, DetailID) → {Month, Year, MessCharges, OtherCharges}
 - Normalization Check:
 - 1NF ✓
 - 2NF ✓
 - 3NF ✓
 - BCNF ✓
- Satisfied, No violation.

FINAL NORMALIZED RELATIONS:

1. WARDEN (wardenID, WardenName, Email, Joining Date)
2. WARDEN - PHONE (WardenID, PhoneNo)
3. HOSTEL (HostelID, HostelName, TotalFloors, WardenID)
4. ROOM - TYPE (Type, Capacity, RoomRent)
5. ROOM (RoomNo, FloorNo, Type, HostelID)
6. STUDENT (StudentID, FirstName, LastName, Gender, DOB, Email, Blood Group, Plantype, MentorStudentID, RoomNo, MessID)
7. STUDENT - PHONE (StudentID, PhoneNo)
8. MESS (MessID, MessName, MessType, Location)
9. MESS - CONTACT (MessID, ContactNo)
10. MEAL (MealID, MealName, Description, Cost, MessID)
11. STAFF (StaffID, StaffName, JoinDate, Salary, Role, MessID)
12. STAFF - PHONE (StaffID, PhoneNo)
13. SUPPLIER (SupplierID, SupplierName)
14. SUPPLIER - PHONE (SupplierID, PhoneNo)
15. INVENTORY - ITEM (ItemID, ItemName, Category, Unit)
16. PROCURES (MessID, SupplierID, ItemID, Quantity)
17. PAYMENT (PaymentID, StudentID, Amount, PaymentMode, Status, PaymentDay, PaymentMonth, PaymentYear)
18. PAYMENT - DETAIL (PaymentID, DetailID, Month, Year, MessCharges, OtherCharges)